

Finite Typed Context Descent

Semantic Visibility, Relational Acyclicity, and Comparison Reconstruction

WuJun Chen

Independent Researcher
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This paper is Paper XXIV of the RIME program. It develops a finite local-to-global theory for typed section and relation data and applies object-appropriate reconstruction gates to fixed-alignment comparison.

Abstract

Problem. Finite structural records are often observed through overlapping typed contexts. Equality on every local context need not imply global equality when the contexts cover labels but omit relational coordinates. Even after all coordinates are covered, local admissibility checks may fail to imply global admissibility when the visible constraints do not semantically entail the full constraint package. Relation-valued local data introduce a different obstruction: pairwise-consistent local relations need not admit a global relation when the context hypergraph is cyclic.

Approach. A context is modeled as a finite typed atom set with definedness-sensitive partial sections. The construction separates label coverage, relation-coordinate coverage, type and convention guards, and overlap compatibility. For a global constraint package Q_C , the visible package Q_U records exactly the constraints whose scopes occur in some patch. Relation-valued descent is treated as a different local object type and is composed with the classical Beeri–Fagin–Maier–Yannakakis acyclicity characterization.

Results. Free sections are separated exactly when every admissible difference support meets the observed coordinates, and a full coordinate cover gives unique gluing. The equalizer of locally admissible sections is the global model space of Q_U , so full constrained descent holds exactly when $Q_U \models_{A_C} Q_C$. Under universal quantification over independently interpreted scope predicates, full admissible descent is equivalent to visibility of every declared scope. For relation-valued local data, universal admissible descent holds exactly when scope visibility is combined with an α -acyclic context hypergraph. These object-appropriate gates yield a fixed-alignment comparison reconstruction theorem. Exact `UNSEEN_SCOPE` and `CYCLIC_CONTEXT_CORE` fixtures separate the two failure modes.

Boundary. The standard separatedness, free-equalizer, and relational acyclicity results are not claimed as new. The contribution is the typed semantic-visibility interface, its composition with the imported relational obstruction, and its propagation into comparison reconstruction. No Grothendieck topology, arbitrary-presheaf descent theorem, causal inference, or universal comparison theorem is claimed.

Notation Table

Symbol	Meaning
$C = (A_C, \tau_C, \theta_C)$	finite typed context
$u_i : C_i \rightarrow C$	canonical restriction
A_i	ambient image of the atoms retained by patch i
O_U	union of all observed atom sets A_i
$F_{\text{free}}(C)$	definedness-sensitive free typed sections on C
G	declared class of global sections
Q_C	global constraint package
Q_U	constraints visible in at least one patch
H_U	context hypergraph with edges A_i
R_i	relation-valued local object on A_i
$\bowtie_i R_i$	natural join of the local relations
Φ, Θ	fixed alignment and comparison-semantics packages

1 Introduction

This paper proves finite statements about explicit typed coordinate sets. It does not claim a Grothendieck topology, a topos-level result, descent for arbitrary presheaves, or a universal comparison theorem. In particular,

$$\begin{aligned}
 &\text{label coverage} \neq \text{relation-coordinate coverage,} \\
 &\text{separatedness} \neq \text{existence,} \\
 &\text{free gluing} \neq \text{constrained gluing,} \\
 &\text{section-valued descent} \neq \text{relation-valued descent,} \\
 &\text{local comparison compatibility} \neq \text{causal attribution.}
 \end{aligned}$$

Theorems 3 and 4 instantiate standard mathematical patterns: joint monicity of restriction families and equalizer gluing for finite coordinate products. They are not claimed as new universal sheaf theorems. The contribution lies in the following combined interface:

1. signatures are typed partial maps whose definedness is part of equality;
2. relation-coordinate coverage is checked independently of label coverage;
3. fixed-package section descent is characterized by semantic entailment from the visible constraint package;
4. uniform section descent is separated from the imported α -acyclic criterion for relation-valued local objects;
5. two AB/BC/AC controls isolate semantic-visibility and cyclic-relation failures;
6. the resulting object-appropriate descent gates are exposed for contextual comparison and finite audit reconstruction without importing causal or action semantics.

The main theorem spine is

Free Section Gluing
→ Exact Admissible-Section Characterization
→ Universal Scope-Visibility Corollary
→ Imported Relational Acyclicity Theorem
→ Typed Relational Admissible-Descent Characterization
→ Comparison Reconstruction.

1.1 Scope and Positioning

This paper studies finite local-to-global conditions for restriction, separatedness, and gluing. Paper XIII supplies the motivating comparison interface [4], but is an application source rather than a premise of the theorems below. Conversely, these descent theorems do not validate or promote a Paper XIII comparison.

This line isolates a recurring structural question. Given an operation $T : X \rightarrow X$ and a lossy map $q : X \rightarrow Y$, an operation on Y represents T only when some well-defined $\bar{T} : Y \rightarrow Y$ makes

$$q \circ T = \bar{T} \circ q.$$

Failure of injectivity alone neither proves nor disproves such descent; the issue is whether the structure is compatible with the information identified by q . Here the maps are finite contextual restrictions, and the theorems state explicit coordinate-coverage and constraint-scope conditions under which separation, gluing, or comparison reconstruction is justified.

2 Typed Contexts and Signatures

2.1 Contexts

Fix a universe of typed atoms. A finite typed context is

$$C = (A_C, \tau_C, \theta_C),$$

where A_C is a finite set of declaration atoms, $\tau_C(a)$ is the payload type of atom a , and θ_C is a fixed convention package. Atoms may represent sector values, observable values, typed incidences, word-support coordinates, or convention guards when those guards vary as section data. Guards fixed by θ_C index the context and are not duplicated as section coordinates.

A canonical restriction $u : C' \rightarrow C$ is an injective type-preserving atom map

$$u_{\#} : A_{C'} \hookrightarrow A_C$$

that preserves the declared convention package. Identity and composition are the corresponding identity and composite injections.

Theorem 2.1 (Finite Typed-Context Category).

Finite typed contexts and canonical restrictions form a category $\mathcal{C}_{\text{ctx}}$.

Proof. Identity atom maps are injective and preserve types and conventions. Composites of injective type-preserving maps have the same properties. Associativity and the identity laws follow from function composition. $\square$

2.2 Free Typed Signatures

Let V_a be the payload set declared by $\tau_C(a)$. A free typed signature section on C is a dependent partial map

$$s \in F_{\text{free}}(C) := \prod_{a \in A_C} \text{Option}(V_a).$$

The value `none` means that the coordinate is not defined in the section; it is not a numerical zero. Restriction along $u : C' \rightarrow C$ is precomposition with $u_{\#}$:

$$F_{\text{free}}(u)(s)(a) = s(u_{\#}a).$$

Theorem 2.2 (Restriction Functor).

The assignment $F_{\text{free}} : \mathcal{C}_{\text{ctx}}^{\text{op}} \rightarrow \mathbf{Set}$ is a contravariant functor.

Proof. Precomposition by an identity atom map is the identity. Precomposition by a composite equals successive precomposition in reverse order. Payload types and convention guards are preserved by the morphism obligations.

This theorem is specific to canonical restriction. Resolution refinement, aggregation, alignment, and convention conversion require separately typed maps and are not silently admitted as context morphisms. $\square$

3 Separating Covers

Let

$$\mathcal{U} = \{u_i : C_i \rightarrow C\}_{i=1}^m$$

be a nonempty finite family of canonical restrictions, with $m \geq 1$. Its observed atom set is

$$O_{\mathcal{U}} = \bigcup_{i=1}^m (u_i)_{\#} A_{C_i} \subseteq A_C.$$

For two sections define their difference support, including defined/undefined differences, by

$$\Delta(s, t) = \{a \in A_C : s(a) \neq t(a)\}.$$

Four checks must remain distinct for any theorem application:

1. **label coverage:** the underlying sector/observable labels are jointly retained;
2. **relation-coordinate coverage:** $O_{\mathcal{U}}$ contains every typed atom on which the relevant global section class may differ;
3. **type/convention guards:** every u_i preserves payload types and the declared convention package, as required by a canonical restriction;
4. **overlap compatibility:** the particular local family agrees in both payload and definedness on every $A_i \cap A_j$.

The first and third properties concern the declared context family, the second concerns its section-separating power, and the fourth concerns a supplied local family. None is inferred from another.

For a declared global section class $G \subseteq F_{\text{free}}(C)$, call $\mathcal{U}$ **G -separating** when the restriction map

$$\rho_{\mathcal{U}} : G \longrightarrow \prod_i F_{\text{free}}(C_i)$$

is injective.

Theorem 3.1 (Separating-Cover Characterization).

A finite context family $\mathcal{U}$ is G -separating if and only if

$$\forall s \neq t \in G, \quad \Delta(s, t) \cap O_{\mathcal{U}} \neq \emptyset.$$

Proof. If the intersection contains a , choose a patch containing a . The two restrictions differ at its local representative, so $\rho_U(s) \neq \rho_U(t)$. Conversely, if a distinct pair has difference support disjoint from O_U , the pair agrees on every atom retained by every patch and therefore has identical restrictions.

This is the necessary-and-sufficient finite criterion for a constrained or candidate-relative section space. Pure coordinate coverage is a stronger criterion that becomes exact for the unrestricted free presheaf. $\square$

Corollary 3.2 (Universal Coordinate Criterion).

Assume each declared payload type V_a is inhabited. The restriction family is separating on all of $F_{\text{free}}(C)$ if and only if

$$O_U = A_C.$$

Proof. Coverage implies that every difference is seen by some patch. If an atom a is uncovered, compare the everywhere-undefined section with the section that has one declared value only at a . They are distinct globally and equal on every patch.

Thus “sector labels cover the ambient sector set” is insufficient unless it also implies coverage of every typed relation coordinate on which global sections may differ. $\square$

4 Free Finite Gluing

For $A_i = (u_i)_\# A_{C_i}$, a local family (s_i) is **exactly matching** when for every i, j its ambiently reindexed partial maps agree on $A_i \cap A_j$. Agreement is equality in **Option** payloads: both values and definedness must agree. Merely agreeing where both sides happen to be defined is weaker and does not constitute a matching family for this theorem.

All overlaps in this paper are ambient typed-atom intersections. Write

$$A_{ij} := A_i \cap A_j,$$

with payload types inherited from A_C , and define

$$F_{\text{free}}(A_i) := \prod_{a \in A_i} \text{Option}(V_a).$$

No categorical pullback $C_i \times_C C_j$ is assumed or used. In particular, the notation below concerns restrictions to A_{ij} , not the existence of an unproved pullback object in $\mathcal{C}_{\text{ctx}}$.

Theorem 4.1 (Finite Free Gluing).

If $O_U = A_C$, every exactly matching family $s_i \in F_{\text{free}}(A_i)$ has a unique global extension $s \in F_{\text{free}}(A_C)$.

Proof. For $a \in A_C$, choose a patch i containing a and define $s(a)$ to be the corresponding value of s_i . Exact overlap compatibility makes the definition independent of the chosen patch. The resulting section restricts to every s_i . If t is another extension, coordinate coverage supplies a patch containing each a , and both $s(a)$ and $t(a)$ equal its local value. Hence $s = t$.

Writing every overlap as the declared ambient intersection A_{ij} , Theorem 4.1 gives

$$F_{\text{free}}(A_C) \cong \text{Eq} \left(\prod_i F_{\text{free}}(A_i) \rightrightarrows \prod_{i,j} F_{\text{free}}(A_{ij}) \right).$$

No separate “local constraint completeness” is needed for the free partial-map presheaf. That obligation appears only after an admissible subspace is imposed. $\square$

5 Exact Admissible-Section Descent

A finite constraint package Q_C consists of constraints q , each with a finite atom scope

$$\text{supp}(q) \subseteq A_C$$

and a predicate depending only on the restriction of a section to that scope. For every atom subset $D \subseteq A_C$, define the induced local constraint package

$$Q_C|_D = \{q \in Q_C : \text{supp}(q) \subseteq D\}.$$

Because each q depends only on its declared scope, it can be evaluated on a section over D whenever $q \in Q_C|_D$. Define the local admissible section space with the constraint package visible on D by

$$F_{Q|_D}(D) = \{s \in F_{\text{free}}(D) : q(s) = \text{true for every } q \in Q_C|_D\}.$$

In particular,

$$F_{Q_C}(A_C) = \{s \in F_{\text{free}}(C) : q(s) = \text{true for every } q \in Q_C\}.$$

For $D' \subseteq D$, restriction sends $F_{Q|_D}(D)$ into $F_{Q|_{D'}}(D')$: every $q \in Q_C|_{D'}$ also lies in $Q_C|_D$, and its value depends only on the unchanged coordinates in $\text{supp}(q)$. Hence the induced local admissible spaces form a restriction-stable admissible subpresheaf of F_{free} on the finite atom-subset category generated by A_C . The explicit notation $F_{Q|_{A_i}}(A_i)$ prevents a local validator from being mistaken for a checker of the full package Q_C .

For a fixed cover $\mathcal{U}$, define the **visible constraint package**

$$Q_{\mathcal{U}} = \bigcup_i Q_C|_{A_i} = \{q \in Q_C : \exists i, \text{supp}(q) \subseteq A_i\}.$$

The corresponding global model space is

$$F_{Q_{\mathcal{U}}}(A_C) = \{s \in F_{\text{free}}(C) : q(s) = \text{true for every } q \in Q_{\mathcal{U}}\}.$$

For constraint packages P, Q over the fixed typed atom set A_C , write

$$P \models_{A_C} Q$$

when every global typed partial assignment satisfying every predicate in P also satisfies every predicate in Q . This is semantic entailment in the fixed typed domains and convention profile; it is not a claim about a chosen syntactic proof calculus.

Equivalently,

$$Q_{\mathcal{U}} \not\models_{A_C} Q_C$$

means that some global typed partial assignment satisfies every visible constraint in $Q_{\mathcal{U}}$ while violating at least one constraint in the full package Q_C .

A cover is **scope-complete** for Q_C when

$$\forall q \in Q_C, \quad \exists i, \quad \text{supp}(q) \subseteq A_i.$$

Theorem 5.1 (Exact Admissible-Section Characterization).

Assume $O_U = A_C$. Canonical restriction and free gluing induce a bijection

$$\text{Eq}\left(\prod_i F_{Q|A_i}(A_i) \rightrightarrows \prod_{i,j} F_{Q|A_i \cap A_j}(A_i \cap A_j)\right) \cong F_{Q_U}(A_C).$$

Consequently, full constrained section descent holds exactly when

$$Q_U \models_{A_C} Q_C.$$

Proof. An element of the equalizer is an exactly matching family (s_i) with each $s_i \in F_{Q|A_i}(A_i)$. Theorem 4.1 supplies its unique free gluing s . If $q \in Q_U$, then $\text{supp}(q) \subseteq A_i$ for some i . Local admissibility gives $q(s_i) = \text{true}$, and scope locality gives $q(s) = q(s_i) = \text{true}$. Hence $s \in F_{Q_U}(A_C)$.

Conversely, let $s \in F_{Q_U}(A_C)$. Its restrictions exactly match. If $q \in Q_C|_{A_i}$, then $q \in Q_U$, so the restriction $s|_{A_i}$ satisfies q . Thus every restriction lies in $F_{Q|A_i}(A_i)$. Restriction and free gluing are inverse by Theorem 4.1, proving the displayed bijection.

Because $Q_U \subseteq Q_C$, one always has $F_{Q_C}(A_C) \subseteq F_{Q_U}(A_C)$. The full constrained restriction map is onto the equalizer exactly when these model spaces are equal, which is exactly the semantic entailment $Q_U \models_{A_C} Q_C$.

Thus scope completeness is sufficient but not necessary for a fixed package: an unseen constraint may still be semantically implied by the visible ones. The syntactic condition becomes exact only after the predicate interpretations are universally quantified. $\square$

Corollary 5.2 (Universal Scope-Visibility Characterization).

Assume each payload type V_a is inhabited and predicates on the declared scopes can be interpreted independently. The following are equivalent:

1. for every interpretation of the declared scope predicates, every exactly matching locally admissible section family has a globally admissible gluing;
2. $\mathcal{U}$ is scope-complete for Q_C .

Proof. Scope completeness gives $Q_U = Q_C$, so Theorem 5.1 proves sufficiency. Conversely, suppose a declared constraint q_* has nonempty scope S_* not contained in any patch. Choose a global partial assignment s_* . Interpret every other predicate as a tautology, and interpret q_* as a proper predicate that rejects $s_*|_{S_*}$ and accepts at least one other scope assignment. Such a second assignment exists because S_* is nonempty and each $\text{Option}(V_a)$ contains both **none** and a declared value. The restrictions of s_* are exactly matching and locally admissible, since no patch can evaluate q_* . Their unique free gluing is s_* , which violates q_* . Universal admissible descent therefore fails. An empty scope cannot be invisible because the nonempty cover has a patch and the empty set is contained in every patch. $\square$

6 Relation-Valued Descent

Section-valued descent glues one compatible local assignment. Context cycles do not obstruct that operation: Theorem 4.1 works for every finite coordinate cover. A different criterion appears only after the local object type changes to a set of assignments.

A **typed local relation** on A_i is a subset

$$R_i \subseteq F_{\text{free}}(A_i).$$

For $D \subseteq A_i$, write

$$\pi_D(R_i) = \{s|_D : s \in R_i\}.$$

A family (R_i) is **pairwise consistent** when

$$\pi_{A_i \cap A_j}(R_i) = \pi_{A_i \cap A_j}(R_j) \quad \text{for all } i, j.$$

It is **globally consistent** when some relation $R \subseteq F_{\text{free}}(A_C)$ has exact projections $\pi_{A_i}(R) = R_i$ for every i . The natural join

$$\bowtie_i R_i = \{s \in F_{\text{free}}(A_C) : s|_{A_i} \in R_i \text{ for every } i\}$$

is a canonical candidate. It is the largest relation satisfying all local membership conditions, but global relations with the same exact projections need not be unique.

Let $H_{\mathcal{U}}$ be the context hypergraph with vertex set A_C and hyperedges A_i . It is **α -acyclic** when it admits a join tree: a tree on the context edges such that, for every atom a , the nodes whose edges contain a form a connected subtree. Equivalently, the hypergraph has an empty GYO reduction.

Imported Theorem 6 (Relational Acyclicity Characterization). The following are equivalent:

1. $H_{\mathcal{U}}$ is α -acyclic;
2. for every choice of finite attribute domains, every pairwise-consistent family of relations on the edges A_i is globally consistent.

Under these conditions, the natural join has exact projections onto every A_i .

This is the classical local-to-global characterization of acyclic database schemes due to Beeri, Fagin, Maier, and Yannakakis [3]. Atserias and Kolaitis extend the same characterization to relations over positive semirings and use a generalized Tseitin construction for the cyclic direction [2]. The present typed partial-map statement is a direct specialization obtained by taking the ordinary attribute domain at a to be $\text{Option}(V_a)$. It is imported mathematical structure, not a novelty claim here.

A local relation is **locally admissible** when

$$R_i \subseteq F_{Q|A_i}(A_i).$$

A global relation is **globally admissible** when $R \subseteq F_{Q_C}(A_C)$.

Theorem 6.1 (Typed Relational Admissible-Descent Characterization).

Assume $O_{\mathcal{U}} = A_C$ and fix a finite registry of declared constraint scopes. The following are equivalent:

1. $H_{\mathcal{U}}$ is α -acyclic and $\mathcal{U}$ is scope-complete;
2. *for every choice of inhabited finite payload types, every independent interpretation of the declared scope predicates, and every pairwise-consistent locally admissible relation family (R_i) , there is a globally admissible relation R satisfying $\pi_{A_i}(R) = R_i$ for all i .*

When these conditions hold, the natural join $\bowtie_i R_i$ is one canonical such global relation.

Proof. Suppose condition 1 holds. Imported Theorem 6 shows that $R^\bowtie = \bowtie_i R_i$ has exact projection R_i on every patch. Let $s \in R^\bowtie$ and $q \in Q_C$. Scope completeness supplies i with $\text{supp}(q) \subseteq A_i$. Since $s|_{A_i} \in R_i$ and R_i is locally admissible, $q(s|_{A_i}) = \text{true}$. Scope locality gives $q(s) = \text{true}$. Thus every tuple of $R^\bowtie$ satisfies every global constraint, so $R^\bowtie \subseteq F_{Q_C}(A_C)$.

Conversely, suppose the universal property in condition 2 holds. If H_U were not α -acyclic, interpret every declared predicate as a tautology. The converse direction of Imported Theorem 6 supplies finite domains and a pairwise-consistent relation family with no global relation, contradicting condition 2.

If scope completeness failed, choose an invisible nonempty scope S_* and a global assignment s_* . Interpret all other predicates as tautologies and let the predicate on S_* reject $s_*|_{S_*}$ while accepting another scope assignment. Set $R_i = \{s_*|_{A_i}\}$ for every i . These singleton relations are pairwise consistent and locally admissible. Coordinate coverage forces every tuple in any relation with exact projections R_i to equal s_* , while exact nonempty projections require that tuple to occur. Such a relation cannot be globally admissible. Hence condition 2 implies scope completeness as well as α -acyclicity.

The two theorem layers must not be collapsed: □

Local object and quantifier	Fixed-instance exact criterion	Uniform structural criterion
one section with constraints	$Q_U \models_{A_C} Q_C$	every declared constraint scope is visible
one local relation per patch	the supplied family is globally consistent/joinable	H_U is α -acyclic

For universal constrained relation descent, the two right-hand criteria occur together. For section-valued descent, hypergraph cyclicity is not an obstruction.

7 Sharp Three-Label Controls

Let the sector labels be A, B, C and let the global relation atoms be

$$A_C = \{AB, BC, AC\},$$

with Boolean payloads.

7.1 Label Coverage Without Separatedness

Take patches retaining the relations AB and BC . Their underlying sector labels jointly cover $\{A, B, C\}$, but AC is unseen. The sections

$$s_0 = (AB = 1, BC = 1, AC = 0), \quad s_1 = (AB = 1, BC = 1, AC = 1)$$

have identical restrictions and are globally distinct. This is a separatedness failure, not an existence failure.

Adding an AC patch gives $O_U = A_C$. Theorems 3 and 4 then give universal separatedness and unique free gluing, without enumerating a global candidate set.

7.2 UNSEEN_SCOPE: Section-Valued Semantic Invisibility

Now impose the ternary even-parity constraint

$$AB \oplus BC \oplus AC = 0.$$

The three relation-coordinate patches retaining AB , BC , and AC cover every relation atom, but no patch contains the full constraint scope. The local assignment

$$AB = BC = AC = 1$$

is overlap-compatible and has a unique free gluing. Its parity is odd, so it has no admissible global extension in $F_{Q_C}(A_C)$. This is an existence failure, not a separatedness failure.

The construction is minimal within the three-coordinate, three-singleton-patch pattern: all coordinates are covered, while one ternary scope crosses every patch. It is not claimed to be minimal among arbitrary presheaves or arbitrary constraint languages. Its conclusion is exactly

$$Q_U \not\models_{A_C} Q_C.$$

7.3 CYCLIC_CONTEXT_CORE: Relation-Valued Nonjoinability

Here AB , BC , and AC denote two-attribute context edges, whereas in the preceding section they denote individual relation-coordinate atoms. The two hostile controls therefore live in different local-object types.

Let the global attribute set be $\{A, B, C\}$, with Boolean domains, and take context edges AB , BC , and AC . Define

$$R_{AB} = \{(0, 0), (1, 1)\}, \quad R_{BC} = \{(0, 0), (1, 1)\},$$

and

$$R_{AC} = \{(0, 1), (1, 0)\}.$$

Every projection onto a unary overlap is the full relation $\{0, 1\}$, so the three local relations are pairwise consistent. A global tuple would have to satisfy

$$A = B, \quad B = C, \quad A \neq C,$$

which is impossible. Hence the family is not globally consistent. This is the minimal three-edge cyclic illustration of Imported Theorem 6; it is not the general counterexample construction for an arbitrary non-alpha-acyclic GYO residual core.

The two hostile controls have different types and cannot substitute for one another:

Control	Local object	Exact failure
unseen scope	one exactly matching local assignment family	a unique free section exists but global admissibility fails
cyclic context	one pairwise-consistent local relation family	no global relation exists

8 Comparison Reconstruction

Let X_R be a declared report-object system with canonical restrictions. It may be section-valued or relation-valued, but the object type is fixed for a given comparison. An $\mathcal{U}$ -**reconstruction gate** is a declared map

$$\text{Rec}_{\mathcal{U}} : \{\text{admissible compatible local } X_R\text{-families}\} \longrightarrow X_R(C)$$

whose result restricts exactly to the supplied local objects. In the section-valued case, Theorem 5.1 supplies this gate when its semantic-entailment condition holds, and the reconstruction is the unique free gluing. In the relation-valued universal case, Theorem 6.1 supplies the natural join as a canonical gate. The latter does not claim that no other global relation has the same local projections.

Let $F_A \subseteq F_{\text{free}}^A$ be a restriction-stable admissible audit-signature subpresheaf. Fix a carrier, an alignment Φ , and a comparison-semantics package Θ . A coordinate-local comparator is a family

$$\text{Comp}_C^{\Phi, \Theta} : X_R(C) \times X_R(C) \rightarrow F_A(C)$$

that is restriction-natural:

$$F_A(u) \text{Comp}_C(r, t) = \text{Comp}_{C'}(X_R(u)r, X_R(u)t).$$

Theorem 8.1 (Finite Comparison Reconstruction).

Let $\mathcal{U}$ be a finite typed context family. Assume:

1. **typed report reconstruction:** declared reconstruction gates produce global reference and target objects $(r, t) \in X_R(C)^2$ with exact local restrictions;
2. **audit separatedness:** $F_A(C) \rightarrow \prod_i F_A(C_i)$ is injective;
3. **comparison naturality:** the fixed- Φ , fixed- Θ comparator commutes with every restriction leg in $\mathcal{U}$.

Then the local comparisons are the restrictions of the unique global audit comparison $\text{Comp}_C^{\Phi, \Theta}(r, t)$.

Proof. The declared reconstruction gates supply r and t with the required local restrictions. Comparison naturality shows that $\text{Comp}_C^{\Phi, \Theta}(r, t)$ restricts to every declared local comparison. If another element of $F_A(C)$ has the same local restrictions, audit separatedness identifies it with that global comparison.

Theorem 5.1 and Theorem 6.1 therefore supply different report-reconstruction gates for different local object types. Theorem 3.1 supplies one finite criterion for audit separatedness when the audit difference supports are hit by the observed coordinates. None of these results automatically supplies a comparator on a new object type: a relation-valued application must separately declare its restriction-natural comparison rule.

This is a fixed-fiber reconstruction statement. It does not infer alignment, choose Θ , identify a defect, or attribute a cause. $\square$

9 Finite Decision Criteria

All active conditions are decidable for explicitly finite inputs.

1. **Universal separatedness:** compute $O_{\mathcal{U}}$ and test $O_{\mathcal{U}} = A_C$.
2. **G -relative separatedness:** for each distinct $s, t \in G$, test whether $\Delta(s, t) \cap O_{\mathcal{U}}$ is nonempty.

3. **Exact matching:** compare `Option` payloads on every pairwise atom intersection, including domains of definition.
4. **Visible package:** compute Q_U by finite scope containment.
5. **Fixed-package admissible descent:** when the typed domains and predicates are finite and executable, enumerate global assignments or invoke the declared finite solver to decide $Q_U \models_{A_C} Q_C$.
6. **Uniform scope visibility:** for every declared scope, test containment in at least one patch.
7. **Relation consistency:** compare exact projections of every pair of finite local relations.
8. **Relational structural gate:** run GYO reduction on H_U ; an empty reduction yields a join-tree certificate, while a nonempty residual yields a structural cyclic-core certificate.
9. **Canonical reconstruction:** use coordinate union for matching sections and natural join for relation families that pass the relevant gate.

For $n = |A_C|$, m patches, q constraint scopes, and a finite candidate space of size g , the universal cover and constraint-scope checks are finite set-containment computations. Candidate-relative separation requires at most $\binom{g}{2}$ difference-support checks. Semantic-entailment complexity depends on how predicates are represented; no representation-independent complexity class is claimed here.

A GYO residual core is a certificate of non-alpha-acyclicity, but it is not by itself the concrete Boolean triangle witness of Section 7.3. Automatically constructing pairwise-consistent relations that are globally inconsistent for an arbitrary residual core requires the general classical counterexample construction. This paper keeps the structural certificate and the minimal three-edge hostile fixture distinct.

10 Related Work and Novelty Boundary

10.1 Sheaf Separatedness and Gluing

In sheaf theory, injectivity into a product of restrictions is the familiar separatedness or uniqueness half of the sheaf condition, while the equalizer diagram expresses compatible-family gluing [8]. Theorems 3 and 4 are finite-coordinate instances of that standard structure. This paper does not claim those abstract patterns as new, and it does not declare the active context families to be a Grothendieck topology.

10.2 Database Lossless Joins and Acyclicity

Relational database theory studies lossless decomposition, join dependencies, and the recovery of a global relation from projections [6, 9]. Beeri, Fagin, Maier, and Yannakakis proved that every pairwise-consistent relation family is globally consistent exactly for acyclic database schemes [3]. Atserias and Kolaitis later established a common positive-semiring generalization of the database and probabilistic local-to-global results [2]. Imported Theorem 6 is the ordinary-relation specialization of that classical line. This paper does not claim α -acyclicity, join trees, GYO reduction, or the general cyclic counterexample construction as new.

The section-valued equalizers of Theorems 4 and 5 are a different object: their elements are matching tuples of individual assignments, not compatible families of relations. The typed presentation makes payload types, partial definedness, convention guards, and the change of local object type explicit.

10.3 Constraint Satisfaction and Local Consistency

Constraint-satisfaction methods distinguish local consistency checks from global satisfiability [5]. For a fixed finite constraint package, Theorem 5.1 identifies the exact condition as semantic entailment $Q_U \models_{A_C} Q_C$, not syntactic scope containment. Scope completeness becomes necessary only under the corollary’s universal and independent predicate-interpretation quantifiers. Thus two constraint packages with the same scope hypergraph may have different fixed-instance descent behavior because an unseen predicate may or may not be entailed by the visible ones.

10.4 Contextuality and Finite Relational Structures

Sheaf-theoretic contextuality also organizes local sections, compatibility, and obstructions to global sections [1]. Finite model theory supplies the broader language of finite relational signatures and local/global definability [7]. The present paper does not identify an audit mismatch with physical or quantum contextuality, nor does it take the event-sheaf or empirical-model semantics as its report object. Its narrower object is a typed partial report/audit signature with source-declared relation coordinates and convention guards.

10.5 Novelty Allocation

Relative to these adjacent theories, the contribution is not a new abstract separatedness theorem or a rediscovery of relational acyclicity. It is the interface-level package consisting of:

- typed partial-map carriers with exact definedness-sensitive overlaps;
- an explicit `label-cover` / `relation-cover` / `type-convention-cover` separation;
- the exact visible-package equalizer and fixed-package criterion $Q_U \models_{A_C} Q_C$;
- a strict quantifier boundary between fixed semantic entailment and uniform syntactic scope visibility;
- typed composition of that visibility gate with the imported relational acyclicity obstruction;
- separate unseen-scope and cyclic-context controls;
- propagation of object-appropriate reconstruction gates into comparison under fixed alignment and semantics.

11 Computational Evidence

Two exact finite controls and a pinned Lean companion accompany the manuscript. The Lean source covers free coordinate coverage, separatedness, free gluing, and the three-label missing-coordinate witness. The Python replay checks the `UNSEEN_SCOPE` and `CYCLIC_CONTEXT_CORE` fixtures as distinct typed objects.

These computations do not prove the general relational acyclicity theorem, the visible-package theorem, or the comparison-reconstruction theorem. The imported BFMY result is supported by its cited proof, not by the three-edge replay. Conversely, the manuscript theorems do not validate a concrete SOF adapter or promote a Paper XIII audit. An application must separately bind the relevant coordinate, scope, relation, and convention registries.

12 Evidence Boundary

The manuscript, Lean companion, exact fixtures, imported literature, and release receipt establish different claim surfaces. They must not be promoted across rows of the following table.

Claim surface	Evidence status	Explicit boundary
Theorem 3.1 and Corollary 3.2: free coverage/separatedness	Lean formalized and locally compiled under the pinned closure	The Lean model uses Option -valued finite signatures; it does not formalize every manuscript convention field
Theorem 4.1: free finite gluing	Lean formalized and locally compiled	Covers exactly matching free local sections; no constraints or relation-valued families
Section 7.1: three-label nonseparation witness	Lean formalized	Covers the missing-AC section witness only
Theorem 5.1 and Corollary 5.2: exact admissible-section and universal scope visibility	Manuscript proofs	Not Lean formalized; the exact theorem is relative to fixed typed domains and the corollary requires independent predicate interpretations
Imported Theorem 6: relational acyclicity	Imported published theorem	Not reproved by Lean or by the paper-owned three-edge fixture
Theorem 6.1: typed relational admissible descent	Manuscript proof plus exact hostile control	The Python fixture verifies one minimal cyclic family, not the general BFMY cyclic construction
Section 7.2 and Section 7.3 hostile controls	Exact finite Python replay	Establish only the declared finite fixtures
Theorem 8.1: comparison reconstruction	Manuscript conditional theorem	No comparator implementation or Paper XIII audit is validated
Release receipt	Source-addressed local closure verification	Binds declared bytes and replay status; it does not independently validate its own implementation

13 Claim Status and Boundary

Claim	Status
finite typed-context category	Theorem
restriction functor	Theorem
G -relative separating-cover characterization	Theorem; standard injectivity pattern, typed finite specialization
universal coordinate-cover characterization	Theorem; finite free-coordinate specialization
free finite gluing and equalizer representation	Theorem; standard gluing pattern, definedness-sensitive specialization
exact admissible-section equalizer	Theorem; semantic-visibility core
fixed-package criterion $Q_U \models_{A_C} Q_C$	Theorem
universal scope-visibility characterization	Theorem under explicit independent-interpretation quantifiers
relation local-to-global iff α -acyclic	Imported classical theorem; not novelty
typed universal admissible-relation descent	Theorem; typed composition with imported relation theorem
fixed- Φ , fixed- Θ comparison reconstruction	Theorem under an object-appropriate reconstruction gate
finite decision procedures	Theorem; constructive consequence of the finite definitions
UNSEEN_SCOPE and CYCLIC_CONTEXT_CORE	Computational Certificate; exact finite hostile witnesses
arbitrary-core relation witness generation	Research Program; not supplied by GYO residue alone
natural Grothendieck topology for application contexts	Research Program
descent under refinement, aggregation, or convention change	Research Program

14 Conclusion

Finite typed context descent has different exact gates for different local object types. For section-valued data, observed difference supports determine separatedness, full coordinate coverage gives unique free gluing, and the visible constraint package determines whether local admissibility descends. For relation-valued data, pairwise consistency additionally requires the classical α -acyclicity gate for a universal local-to-global guarantee.

These distinctions propagate directly into comparison reconstruction. A fixed-alignment comparator descends only after the report objects have an object-appropriate reconstruction gate and the audit signature is separating. Local agreement alone does not supply missing relation coordinates, global admissibility, alignment, defect attribution, or causal interpretation.

15 Outlook

Open problems include formalizing the visible-package equalizer, producing constructive witnesses from arbitrary cyclic GYO cores, and characterizing which derived-summary maps preserve separatedness. Minimum semantic-descent covers and sharp complexity bounds depend on the chosen predicate representation. These questions remain within local-to-global mathematics and do not require action, authorization, or causal semantics.

Appendix A Computational Artifacts

All listed artifacts are available in the [RIME repository](#) under `experiments/paper24/`; paths below are relative to that directory.

Artifact	Role	Short path
formal source and closure	finite theorem surface with pinned Lean/Mathlib inputs	<code>lean/</code>
exact fixture records	section- and relation-valued controls	<code>results/</code>
validators	deterministic replay and local closure verification	<code>validation/</code>
release manifest	declared acyclic artifact closure	<code>release-manifest.json</code>

From the repository root, run:

```
python experiments/paper24/validation/validate_release.py
```

The receipt is excluded from its own closure. A passing local validator binds the declared bytes and replay results; it is not independent mathematical validation.

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